Chapter (28):-

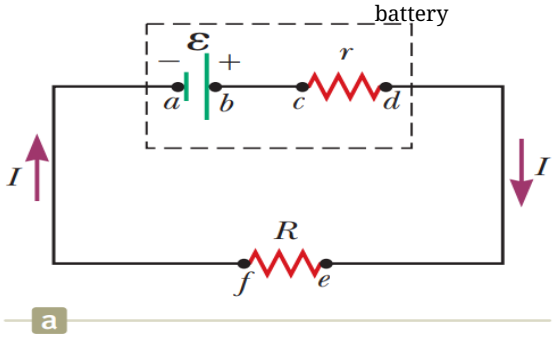
Direct-current Circuits:-

28.1 Electromotive force:-

battery produces a current in a circuit so it's called ↑ a source of electromotive force (emf) ($\mathcal{E}$).

as emf is constant the current is constant too which is called direct current (DC).

electromotive force ($\mathcal{E}$) is :- the maximum possible voltage the battery can provide between it's terminals.

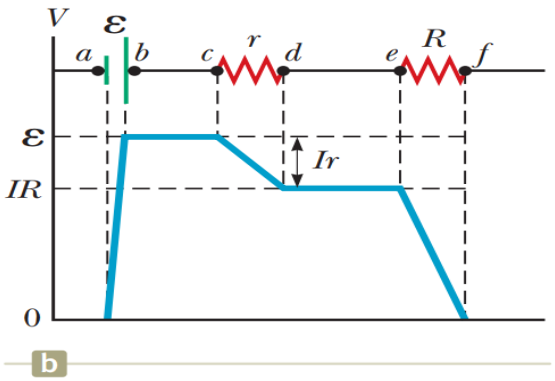


a

battery moves charges from Low to high voltage across it's terminals, and from high to Low voltage across the circuit.

r:- internal resistance of the battery.

as the charge moving from a→b Potential difference increases by $\mathcal{E}$.



b

moving through r reduces the potential difference by Ir. $\Rightarrow$ terminal voltage for the battery (ΔV_{ad})

$$\Delta V_{ad} = \mathcal{E} - Ir$$

R :- is the Load resistance

the potential difference across $R = IR = \Delta V_{ad}$

$$\mathcal{E} - Ir = IR$$

$$\mathcal{E} = Ir + IR$$

$$\boxed{I = \frac{\mathcal{E}}{r+R}}$$

$$[\mathcal{E} = Ir + IR] * I$$

$$I\mathcal{E} = I^2 r + I^2 R$$

the total power associated by emf ($I\mathcal{E}$) is delivered to the load resistance by I^2R and battery internal resistance by I^2r.

$\Delta V_{ad} = \mathcal{E}$ if, 1- $I = 0$, open circuit

2- ideal battery ($r = 0$)

Ex(28.1):- Terminal Voltage of a battery:-

emf = 12 V $r = 0.05\ \Omega$ $R = 3\ \Omega$

a- $I = \frac{\mathcal{E}}{r+R} = \frac{12}{3+0.05} = 3.93\ A$

terminal voltage of the battery $\Delta V = \mathcal{E} - Ir$

$\Delta V = 12 - 3.93 \times 0.05$

$= 11.8$ volt

$\Delta V = IR = 3.93 \times 3 = 11.8\ V$

b- $P_R = I^2 R = (3.93)^2 \times 3 = 46.3$ watt

$P_r = I^2 r = (3.93)^2 \times 0.05 = 0.772$ watt

$P = P_R + P_r = 46.3 + 0.772$

$= 47.1$ watt.

what if?? $r = 2\ \Omega$ for old battery

$I = \frac{\mathcal{E}}{r+R} = \frac{12}{3+2} = 2.4\ A$

$\Delta V = \mathcal{E} - Ir = 12 - 2.4 \times 2 = 7.2$ volt

$P_R = I^2 R = 2.4^2 \times 3 = 17.3$ watt

$P_r = I^2 r = 2.4^2 \times 2 = 11.5$ watt

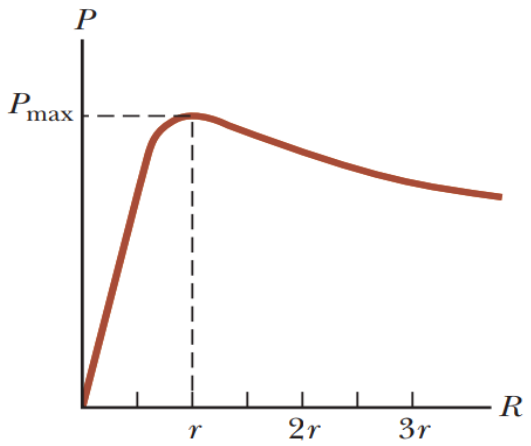
∴ increasing (r) reduces battery ability to deliver energy to R.

Ex(28.2):- Matching the Load:-

$$P_R = I^2 R$$

$$I = \frac{\mathcal{E}}{r+R} \Rightarrow I^2 = \frac{\mathcal{E}^2}{(r+R)^2}$$

$$P_R = \frac{\mathcal{E}^2 R}{(R+r)^2}$$



For maximum power $\frac{dP}{dR} = 0$

$$\frac{d}{dR}\left[\frac{\mathcal{E}^2 R}{(R+r)^2}\right] = 0$$

$$\frac{\mathcal{E}^2(R+r)^2 - 2R\mathcal{E}^2(R+r)}{(R+r)^4} = 0$$

$$\frac{1}{(R+r)^2} - \frac{2R}{(R+r)^3} = 0$$

$$\frac{1}{(R+r)^2} = \frac{2R}{(R+r)^3} \Rightarrow R+r = 2R$$

$$\Rightarrow R = r$$

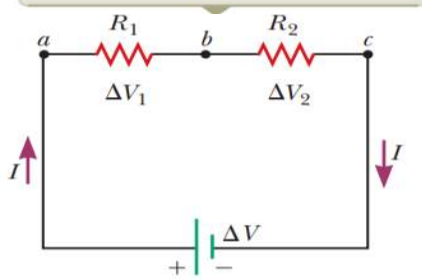
$$P_{max} = \frac{\mathcal{E}^2 r}{(r+r)^2} = \frac{\mathcal{E}^2 r}{4r^2}$$

$$\boxed{P_{max} = \frac{\mathcal{E}^2}{4r}}$$

28.2 Resistors in Series and Parallel:-

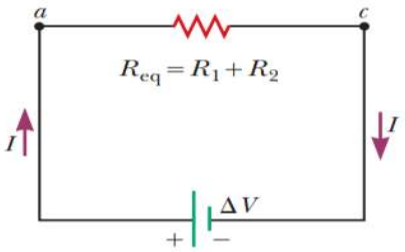
i) Series combination:-

A circuit diagram showing the two resistors connected in series to a battery



b

A circuit diagram showing the equivalent resistance of the resistors in series



c

$$I = I_1 = I_2$$

$$\Delta V = \Delta V_1 + \Delta V_2$$

$$I R_{eq} = I_1 R_1 + I_2 R_2$$

$$R_{eq} = R_1 + R_2$$

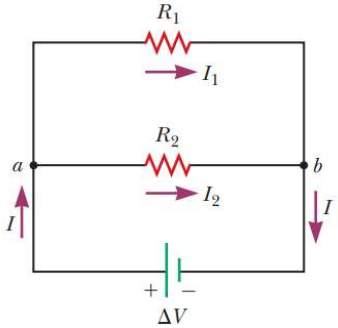
for more than 2 resistors connected in series

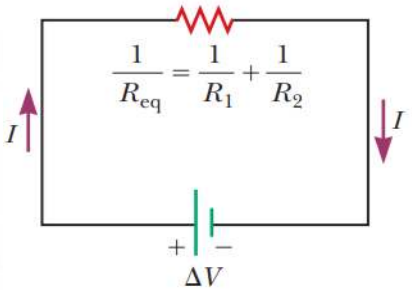
$$R_{eq} = R_1 + R_2 + R_3 + \cdots$$

R_{eq} > individual resistors.

if one resistor of those connected in series creates an open circuit, all resistors are inoperative.

2) Parallel combination





for parallel combination

$$\Delta V = \Delta V_1 = \Delta V_2$$

$$I = I_1 + I_2$$

$$\frac{\Delta V}{R_{eq}} = \frac{\Delta V_1}{R_1} + \frac{\Delta V_2}{R_2}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

for more than 2 resistance in parallel combination

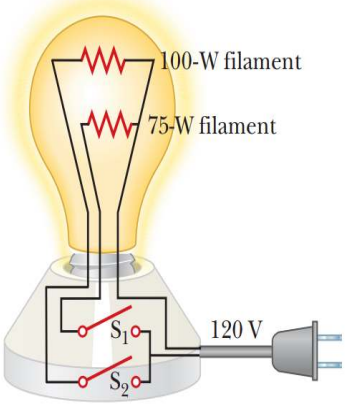
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots$$

R_{eq} < smallest resistance in the group.

in house hold circuits the appliances are connected in parallel; if one device switched off, the others remain on.

Examples of practical applications of series and Parallel circuits:-

1- this figure shows 3-levels of light intensity, using Lamp with 2-filaments. When S_1 closed only current will operate 75 watt-filament, when S_2 closed only current will operate



100-watt filament and when both switches are closed, the current will operate both filaments with 175 watt Power as they connected in parallel. but if they connected in series and one of them breaks the another one will not work.

2- strings of light that are used in decorating. what if we use parallel or series combination for those strings?

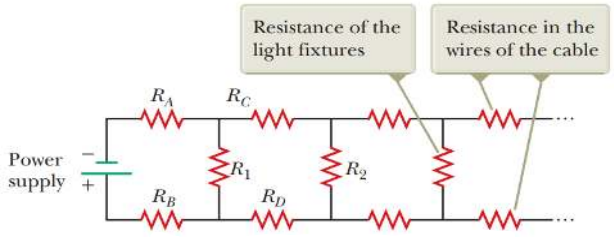
Series combination	Parallel combination
- less operating energy per bulb.	- higher operating energy Per bulb.
- less glowing bulbs	- brighter bulbs.
- lower temperature bulbs	- higher temperature bulbs
- more safe inside home	- more dangours.
- if single filament failed all the rest will go out	- single filament failed will not affect the rest.

To use benifits of both ways of combination, anew design was developed (miniature lights in series).

as one bulb breaks it's represent higher resistance, must of ΔV is applied across it. inside each light bulb in this connection a small jumper loop covered by an isulator, when the filament fails an arc burns the insulator and connects the filament leads which is complete the circuits and light the rest of light bulbs. This reduces the Req which increases current in each bulb and makes the light bulbs glower with higher temperature, which will reduce the lifetime of the bulb. You should ~~reduce~~ change the break bulb to give the rest longer life.

EX (28.3):- land scape Lights:-

the resistance with letter subscripts represent the resistance of wires, and resistors with number subscripts represents the resistance of the light.



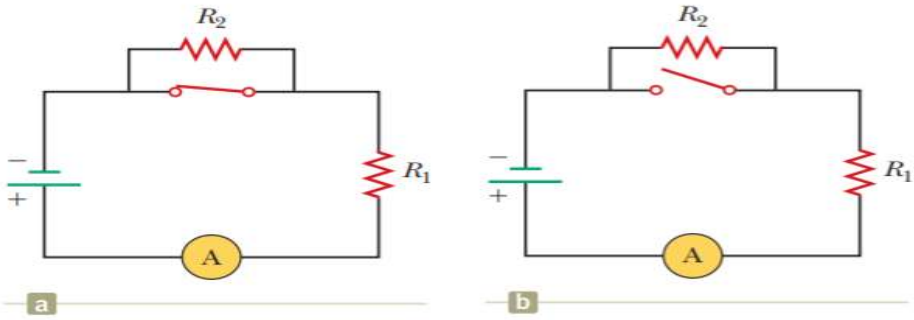
Part of ΔV is dropped due to R_A and R_B, so the voltage across R_1 is less than ΔV, more voltage drop across R_C and R_D $\Rightarrow$ ΔV across R_2 < ΔV across R_1

this make each successive light has a smaller voltage across it and glow less brightly than the one before.

quick quiz (1) Page 834:-

to maximize power that delivered to R, the internal resistance should be (a) as low as Possible

quick quiz (2) Page 837:-



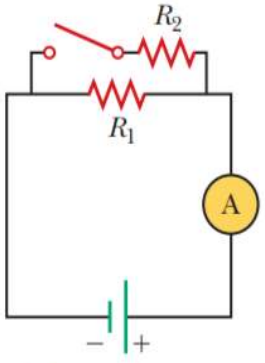
what happens to the reading of the ammeter when switch is opened?

(b)- the reading goes down.

quick quiz (3) Page (840):-

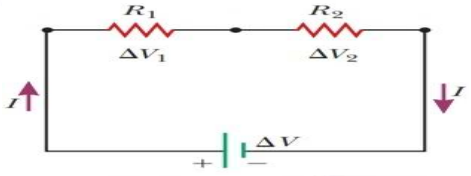
if the switch is closed, what happens to the reading of ammeter?

(a)- the reading increases.



quick quiz (4) page (840):-

i- a third resistor is added in series what happens to the current?

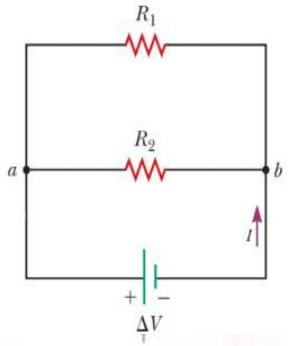


(b)- current decreases

ii- what happens to the terminal voltage?

(a)- terminal voltage increases.

iii- a third resistor is added in parallel, what happens to the current in battery?



(a)- current in battery increases.

iv- what happens to the terminal voltage of battery?

(b)- terminal voltage decreases.

Ex(28.4):- Find the equivalent resistance:-

a- 6 Ω and 3 Ω are connected in parallel

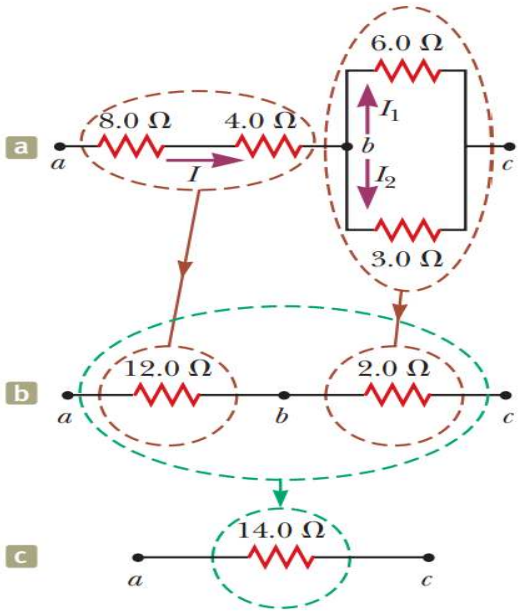
$\frac{1}{R_p} = \frac{1}{6} + \frac{1}{3} \Rightarrow R_p = 2\ \Omega$

8 Ω and 4 Ω are connected in series

$R_s = 8 + 4 = 12\ \Omega$

2 Ω and 12 Ω are in series

$R_{eq} = 2 + 12 = 14\ \Omega$



b - $\Delta V_{ab} = 42$ volt

I passes through 8Ω and 4Ω as they connected in series which is = I passes through $R_{eq} = 14\,\Omega$

$$I = \frac{\Delta V_{ab}}{R_{eq}} = \frac{42}{14} = 3A$$

6Ω and 3Ω are connected in parallel so they have the same voltage difference

$$\Delta V_{6\Omega} = \Delta V_{3\Omega}$$

$$6I_1 = 3I_2 \Rightarrow I_2 = 2I_1$$

but $I = 3A = I_1 + I_2$

$$3 = I_1 + 2I_1 \Rightarrow I_1 = 1A$$

$$I_2 = 2A$$

C- $\Delta V_{8\Omega} = IR = 3 \times 8 = 24$ volt

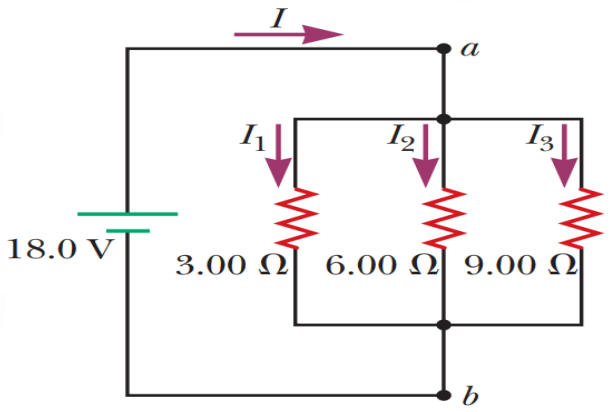
$\Delta V_{4\Omega} = 3 \times 4 = 12$ volt

$\Delta V_{6\Omega} = \Delta V_{3\Omega} = 42 - 24 - 12$

$= 6$ volt

EX(28.5):- Three resistors in parallel:-

a- 3Ω, 6Ω and 9Ω are connected in parallel



$$\frac{1}{R_{eq}} = \frac{1}{3} + \frac{1}{6} + \frac{1}{9}$$

$$= \frac{6+3+2}{18}$$

$$R_{eq} = \frac{18}{11} = 1.64\ \Omega$$

b- $I_1 = \frac{\Delta V}{R_1} = \frac{18}{3} = 6A$

$I_2 = \frac{\Delta V}{R_2} = \frac{18}{6} = 3A$

$I_3 = \frac{\Delta V}{R_3} = \frac{18}{9} = 2A$

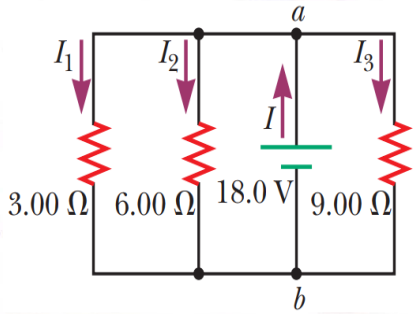
c- $P_1 = I_1^2 R_1 = 6^2 \times 3 = 108$ watt

$P_2 = I_2^2 R_2 = 3^2 \times 6 = 54\ W$

$P_3 = I_3^2 R_3 = 2^2 \times 9 = 36\ W$

$P = P_1 + P_2 + P_3 = 198\ W = \frac{(\Delta V)^2}{R_{eq}} = \frac{18^2}{1.64}$

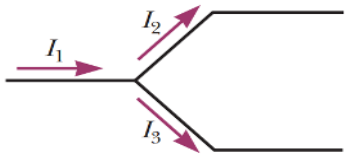
what if ?? there is no effect to the calculations as $\Delta V = 18V$ and resistors are still in parallel.



28.3 Kirchhoff's Rules:-

1- Junction rule:- at any junction, the sum of currents must = zero.

$$\sum_{\text{junction}} I = 0$$



$I_1 - I_2 - I_3 = 0$

$I_1 = I_2 + I_3$

currents directed into junction are +ve, while currents directed out of junction are -ve.

2- Loop rule:- the sum of potential differences across all elements around any closed loop must be zero.

$$\sum_{\text{closed loop}} \Delta V = 0$$

from junction rule:- the electric charge is conserved

$$\sum Q_{in} = \sum Q_{out}$$

from loop rule: the energy is conserved, for moving charge in a closed loop, the sum of increases of energy as a charge passes through elements = the sum of decreases of energy as a charge passes through another elements.

in the figures to the right, the traveling through the circuit elements to the right.

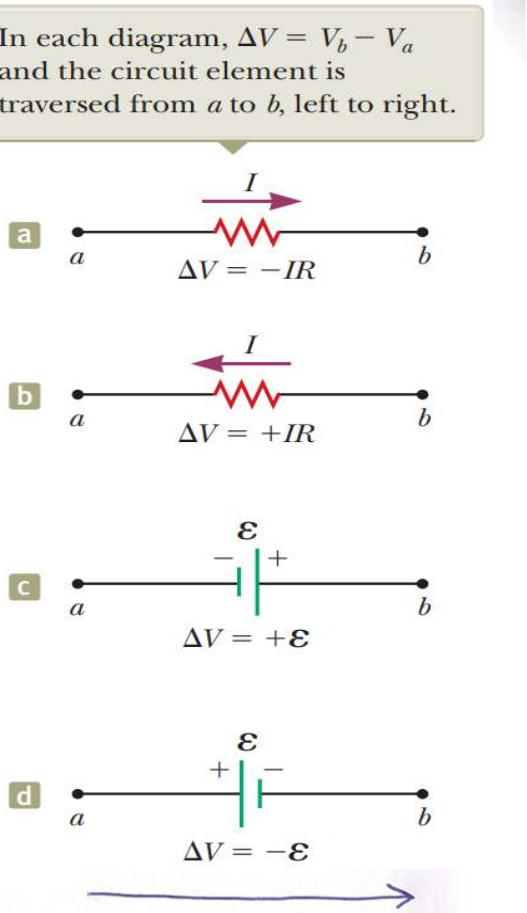
$\Delta V = V_b - V_a$.

ⓐ $\Delta V = -IR$, $V_b < V_a$

ⓑ $\Delta V = IR$, $V_b > V_a$

ⓒ $\Delta V = \mathcal{E}$, $V_b > V_a$

ⓓ $\Delta V = -\mathcal{E}$, $V_b < V_a$



Problem-solving strategy:-

1- determine all elements in the circuit, identify the polarity of each battery and imagine the directions of current in each battery.

2- reduce the circuit by means of combination in parallel and series.

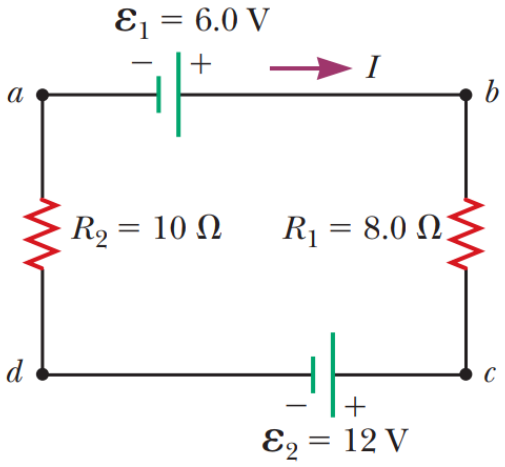
3- assign directions to the current in each Part of the circuit.

4- apply the junction rule to the junction in the circuit, and apply the loop rule to all loops in the circuit after choosing the direction of travel around the loops.

5 - solve equations for unknown quantities.

6- if any of the resulting currents have a -ve value, that means you have guessed incorrect direction for the current but the value is correct.

Ex(28.6):- A single-loop circuit:-



$\sum \Delta V = 0$

closed loop

$\mathcal{E}_1 - I R_1 - \mathcal{E}_2 - I R_2 = 0$

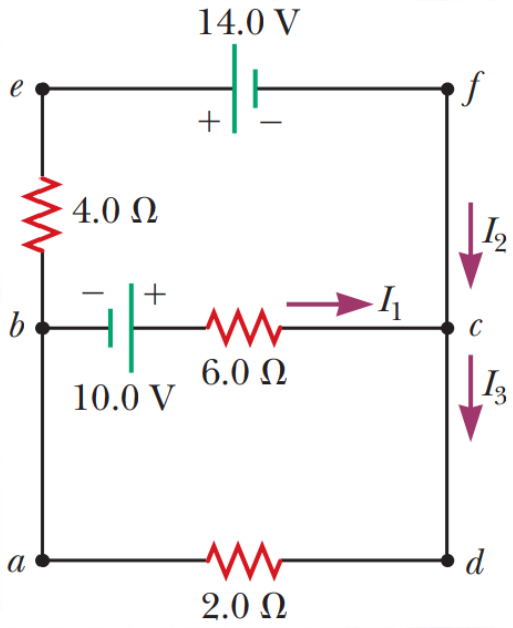
$6 - 8I - 12 - 10I = 0$

$-6 - 18I = 0$

$I = -0.33$ A, the direction of the current is opposite to the assumed one.

Ex(28.7):- A multiloop Circuit:-

$\sum_{\text{junction c}} I = 0$



$I_1 + I_2 - I_3 = 0$ ----①

$\sum \Delta V = 0$ For closed loop (abcdea):-

$10 - 6I_1 - 2I_3 = 0$ ---②

For closed loop (befcb):-

$-4I_2 - 14 + 6I_1 - 10 = 0$

$-24 + 6I_1 - 4I_2 = 0$ ---③

from eq ① $I_3 = I_1 + I_2$, substitute I_3 into eq (2)

$10 - 6I_1 - 2(I_1 + I_2) = 0$

$10 - 8I_1 - 2I_2 = 0$ ---②'

multiply eq(3) by 4 and eq(2)' by 3 then add the results

$$\begin{array}{r} -96 + 24I_1 - 16I_2 = 0 \\ + \quad 30 - 24I_1 + 6I_2 = 0 \\ \hline -66 - 22I_2 = 0 \end{array}$$

$\Rightarrow I_2 = -3\,A$ (in opposite dir)

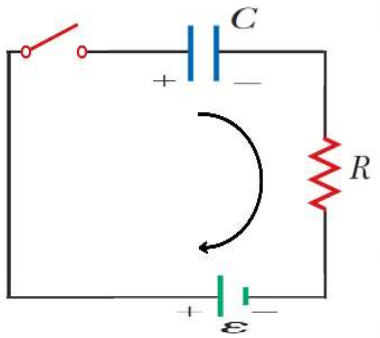
Put I_2 in eq (3) $\Rightarrow$

$-24 + 6I_1 + 12 = 0 \Rightarrow I_1 = 2\,A$

$I_3 = I_1 + I_2 = 2 - 3 = -1\,A$ (in opposite dir)

28.4 RC Circuits:-

a circuit containing a series combination of a resistor and a capacitor is called an RC circuit.



1- Charging a capacitor:-

when S is open, Q on the capacitor equall zero, closing S makes C starts to charge increasing it's Potential difference until C completely full charged it's potential difference = $\mathcal{E}$ then current in R becomes zero.

$$\mathcal{E} - \frac{q}{C} - IR = 0$$

at $t = 0$, $q = 0 \Rightarrow \mathcal{E} = IR$

$$I_o = \frac{\mathcal{E}}{R} \quad (\text{max current when } t = 0)$$

when the capacitor is charged to the max Q the current is zero

$$\mathcal{E} - \frac{q}{C} = 0 \Rightarrow q_{max} = C\mathcal{E}$$

to determine analytical expressions for time dependence of the charge and current we solve eq $\mathcal{E} = \frac{q}{C} + IR$

$$IR = \varepsilon - \frac{q}{C}$$

$$I = \frac{\varepsilon}{R} - \frac{q}{RC}$$

$$\frac{dq}{dt} = \frac{\varepsilon}{R} - \frac{q}{RC}$$

$$= \frac{\varepsilon C - q}{RC} = -\frac{(q - \varepsilon C)}{RC}$$

$$\frac{dq}{q - \varepsilon C} = -\frac{dt}{RC}$$

$$\int_0^q \frac{dq}{q - \varepsilon C} = -\int_0^t \frac{dt}{RC}$$

$$\ln\left(\frac{q - \varepsilon C}{-\varepsilon C}\right) = -\frac{t}{RC}$$

$$\frac{q - \varepsilon C}{-\varepsilon C} = e^{-t/RC}$$

$$q - \varepsilon C = -\varepsilon C\, e^{-t/RC}$$

$$\boxed{q(t) = \varepsilon C\left[1 - e^{-t/RC}\right]}$$

Charging current $I(t) = \frac{dq}{dt}$

$$I(t) = \frac{Q}{RC} e^{-t/RC}$$

$$= \frac{\mathcal{E}}{R} e^{-t/RC}$$

$$\boxed{I(t) = I_0 e^{-t/RC}}$$

the quantity RC is called the time constant (τ)

$$\tau = RC = \left[\frac{\Delta V}{I} \cdot \frac{Q}{\Delta V}\right] = \left[\frac{Q}{\frac{Q}{t}}\right] = [\text{time}] \equiv [\text{sec}]$$

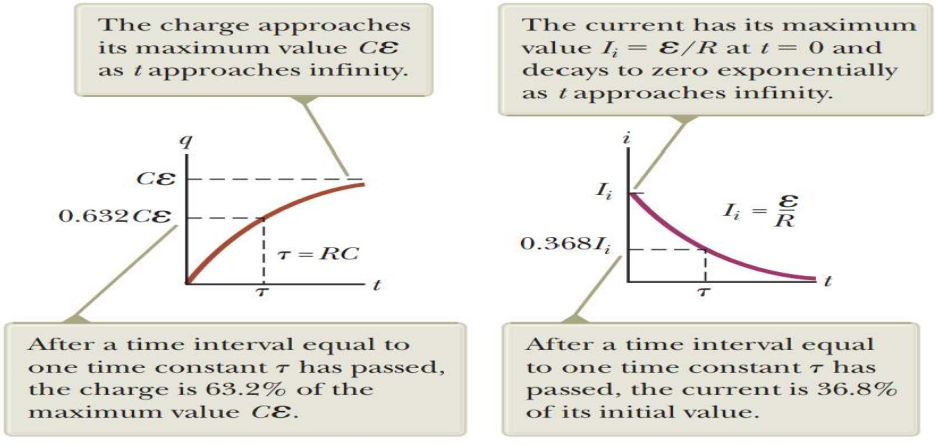
at $t = \tau$

$$q(\tau) = Q_{max}\,(1 - e^{-1}) = 0.632\,Q_{max} = 0.632\,C\mathcal{E}$$

$$I(\tau) = I_0 e^{-1} = 0.368\,I_0 = 0.368\,\frac{\mathcal{E}}{R}$$

τ represent the time interval during which current decreasing to e^{-1} of it's initial value.

after τ passed, the current is 36.8 % of I_0 and the charge is 63.2 of Q_{max}



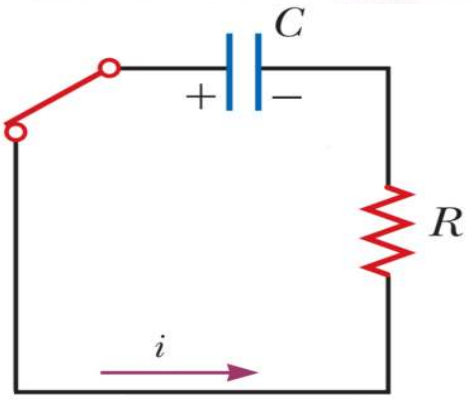
the energy of battery during charging time

$$E = Q_{max}\mathcal{E} = C\mathcal{E}^2$$

half of this energy is stored in the capaeitor and the remaining half stored in the resistance as internal energy.

$$E_C = \frac{1}{2} Q_{max}\mathcal{E} = \frac{1}{2} C\mathcal{E}^2 = E_R$$

2- Discharging a capaeitor:-



for charged C , $\Delta V_c = \frac{Q_{max}}{C}$

$\Delta V_R = 0$ because $I = 0$

if S closed, at $t=0$, Q becomes discharges through R

$$-\frac{q}{C} - IR = 0$$

$$-R\frac{dq}{dt} = \frac{q}{C}$$

$$\frac{dq}{q} = -\frac{dt}{RC}$$

$$\int_{Q_{max}}^{q} \frac{dq}{q} = \frac{-1}{RC}\int_0^t dt$$

$$\ln\left(\frac{q}{Q_{max}}\right) = -\frac{t}{RC}$$

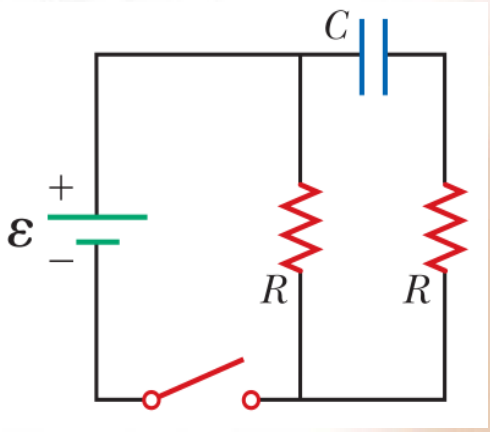
$$\frac{q(t)}{Q_{max}} = e^{-t/RC}$$

$$\boxed{q(t) = Q_{max}\, e^{-t/RC}}$$

$$I = \frac{dq}{dt} \Rightarrow I(t) = \frac{-Q}{RC}\, e^{-t/RC}$$

$$\boxed{I(t) = -I_o\, e^{-t/RC}}$$ the -ve sign indicates that the direction of charging current is opposite to the direction of discharging current.

Quick quiz (5) Page 849 :-



i- just after S is closed.

$$I = \frac{\mathcal{E}}{R_{eq}} \quad , \quad \frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{R}$$

$$= \frac{2}{R}$$

$$R_{eq} = \frac{R}{2} \Rightarrow I_o = \frac{\mathcal{E}}{\frac{R}{2}} = \frac{2\mathcal{E}}{R} \quad (c)$$

ii- after very long time, C is full charge

$$I = \frac{\mathcal{E}}{R} \quad (d)$$

Ex(28.9):- Charging a capacitor in an RC Circuit

$\mathcal{E} = 12\ V \qquad C = 5\ \mu f \qquad R = 8\times10^{5}\ \Omega$

a- $\tau = RC = 8\times10^{5} \times 5\times10^{-6} = 4\ sec$

b- $Q_{max} = C\mathcal{E} = 5\times10^{-6} \times 12 = 60\ \mu C$

c- $I_{max} = I_o = \frac{\mathcal{E}}{R} = \frac{12}{8\times10^{5}} = 15\ \mu A$

d- $q(t) = Q_{max}(1 - e^{-t/\tau})$

$= 60\times10^{-6}(1 - e^{-t/4})$

$I(t) = I_o e^{-t/\tau}$

$= 15\times10^{-6}\ e^{-t/4}$

Ex(28.10):- Discharging a capacitor in an RC circuit

a- $q(t) = \frac{Q_{max}}{4}$

$\frac{Q_{max}}{4} = Q_{max}\, e^{-t/RC}$

$\frac{1}{4} = e^{-t/\tau}$

$\ln(\frac{1}{4}) = \frac{-t}{\tau}$

$-\ln 4 = -\frac{t}{\tau}$

$t = \tau \ln 4 = 1.39\,\tau$

b- $U(t) = \frac{q^2(t)}{2C}$

$= \frac{Q_{max}^2}{2C} e^{-2t/\tau}$

$U(t) = \frac{1}{4} \frac{Q_{max}^2}{2C} = \frac{Q_{max}^2}{2C} e^{-2t/\tau}$

$\ln(\frac{1}{4}) = \frac{-2t}{\tau}$

$-\ln 4 = \frac{-2t}{\tau} \Rightarrow t = \frac{\tau}{2} \ln 4$

$t = 0.693\,\tau$

what if?? half life $t_{1/2}$: is the time required for the charge $q(t)$ to become $\frac{1}{2} Q_{max}$

$\frac{Q_{max}}{2} = Q_{max} e^{-t_{1/2}/\tau}$

$-\ln 2 = -\frac{t_{1/2}}{\tau}$

$t_{1/2} = \tau \ln 2 = 0.693\,\tau$

Ex(28.11):- Energy delivered to a resistor :-

$C = 5\,\mu f \qquad \Delta V = 800 V$

$E_R = \frac{1}{2} C \mathcal{E}^2$

$= \frac{1}{2} \times 5 \times 10^{-6} \times (800)^2$

$= 1.6\ J$